One Shot Is All You Need. Or Is It?

Few-Shot SmolVLA Adaptation on LIBERO

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HSE University · August 2026[1]

GitHub | Hugging Face Models | LIBERO Dataset

1 Question and setup

How many demonstrations does SmolVLA need for a new task—and what does it forget while adapting?

I study SmolVLA, a compact vision-language-action policy [1], on the LIBERO lifelong manipulation benchmark [2]. The setup is intentionally simple:

- **Seen data:** all available `libero_90`: 3,921 episodes, 569,249 frames, and 73 merged instructions.
- **New tasks:** open the middle drawer, put the bowl on the stove, and put the wine bottle on top of the cabinet.
- **Updated weights:** Action Expert and state/action projections; the vision encoder and VLM backbone stay frozen.
- **Target metric:** binary simulator success over fixed rollouts.
- **Retention metric:** the same metric on three frozen `libero_90` probes.

I use the video-backed LeRobot conversion `nvidia/LIBERO_LeRobot_v3` [3, 4]. Exact revisions, episodes, seeds, statistics, and optimizer settings are in Appendix A.

2 First, build a working seen policy

Starting from `lerobot/smolvla_base`, I train the allowed 99.9M parameters for 100k steps. The checkpoint is chosen only on frozen seen probes, before looking at target success.

Checkpoint	60k	80k	**100k**
Seen probes	70.0% (21/30)	76.7% (23/30)	**80.0% (24/30)**

Table 1. Seen-only checkpoint selection.

The 100k checkpoint has final loss 0.23494 and weights SHA-256 prefix `2cd510a594a8`. The training trace is in Appendix C.

3 Then, test zero-shot transfer

The frozen policy almost does not solve the held-out tasks: **1.7% (1/60)**. A paired language control uses the same initial states with correct and wrong instructions. Correct language gives 1.7% (1/60); wrong language gives 0.0% (0/60). Success is too sparse for a strong claim, but action sequences change: mean paired L_2 distances are 0.582, 0.998, and 0.966 for Drawer, Bowl, and Wine. Appendix D.1 gives the full control.

The seen policy works on its training suite, but it does not transfer to the new tasks without adaptation. The next question is how quickly one new skill can be acquired.

[1] This work was prepared for the T-Lab 2026 VLA adaptation assignment.

4 One demonstration changes the problem

The assignment first asks for $N \in \{0, 5, 10, 25\}$. Since $N = 5$ was near the ceiling, I used the allowed extension and added $N = 1$ and $N = 2$. Each cell uses the first N demos, train seeds 42 and 123, and 20 fixed rollouts per seed. No demo is selected by success.

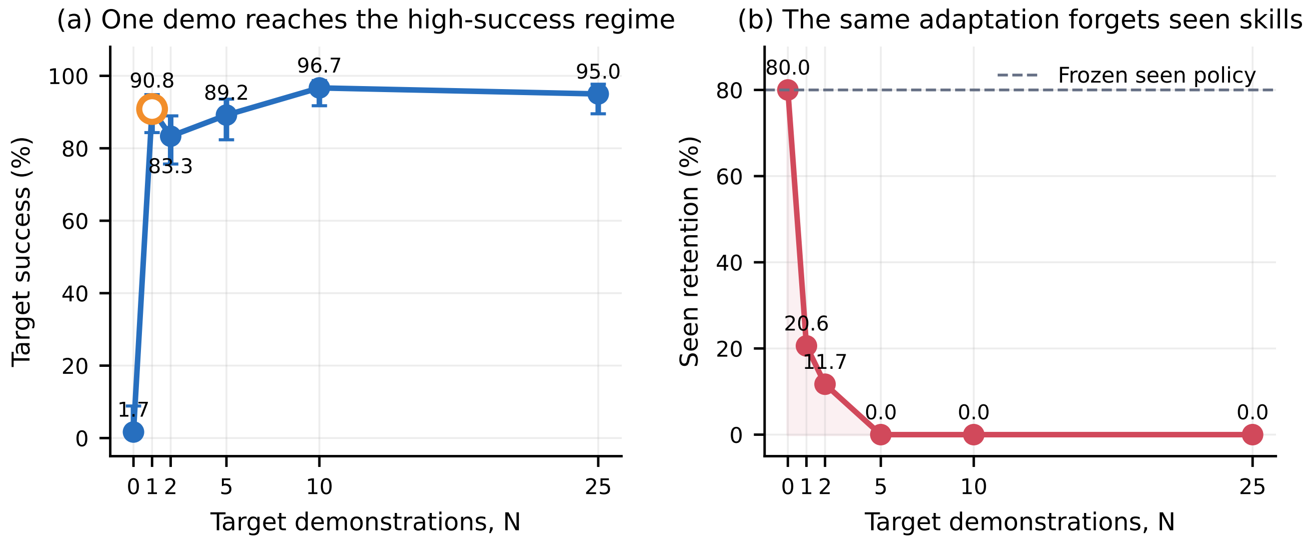


Figure 1. The naive target cost curve and corrected seen retention. Target error bars are 95% Wilson intervals.

The main effect appears after one demonstration: **90.8% (109/120)** target success, versus 1.7% (1/60) at zero shot. Ten demos give the best observed result, 96.7% (116/120), but one shot is already in the high-success regime. $N = 2$ is lower than $N = 1$, so the curve is not monotonic with two seeds.

N	Target success	Seen retention
0	1.7% (1/60)	80.0% (24/30)
1	**90.8% (109/120)**	20.6% (37/180)
2	83.3% (100/120)	11.7% (21/180)
5	89.2% (107/120)	0.0% (0/180)
10	96.7% (116/120)	0.0% (0/180)
25	95.0% (114/120)	0.0% (0/180)

Table 2. Naive adaptation. Exact cells are in Appendix B.

5 But what happened to the old skills?

Figure 1b gives the second result. The frozen policy starts at 80.0% on seen probes. After one target demo, retention falls to 20.6%; at $N \geq 5$, it is 0.0%. High target success and low retention happen together. The first retention run incorrectly used target-overlay statistics and gave 0.0% (0/900). A 2-by-2 control showed that these statistics also break the frozen seen policy. I therefore reran retention with original `libero_90` statistics. Appendix D.2 keeps the control and the rejected result.

One demo is enough to learn the new task, but naive adaptation largely overwrites the old policy. Task 2 is now a stability question.

6 Task 2: what can preserve the seen policy?

6.1 Parameter-efficient updates do not help

Target-LoRA places rank-64 adapters in the Action Expert and keeps projections trainable [5]. Replay-LoRA adds a 25% stream sampled only from `libero_90`; it uses no other target demos and replay is not selected by the probes.

Method	Target success	Seen retention	Trainable
Naive	**90.8% (109/120)**	**20.6% (37/180)**	99.9M
Target-LoRA	82.5% (99/120)	10.6% (19/180)	4.22M
Replay-LoRA	55.8% (67/120)	1.1% (2/180)	4.22M

Table 3. The first preservation ideas. Parameter efficiency is not preservation.

Target-LoRA reduces trainable parameters 23.7 times, but both metrics get worse. Replay-LoRA also mixes actions under incompatible target-only normalization; Drawer loss explodes above 10^{10}. Details are in Appendix C.3.

6.2 Does direct weight anchoring help?

LoRA suggests a different hypothesis: forgetting may depend more on *where* the Action Expert moves than on how many parameters move. Frozen-Stats FT uses canonical `libero_90` statistics with no extra penalty. Anchored FT adds L2-SP [6]:

$$\mathcal{L} = \mathcal{L}_{\text{target}} + \lambda \sum_{\theta_i \in \Theta_{\text{train}}} \|\theta_i - \theta_{i,\text{seen}}\|_2^2, \qquad \lambda = 0.01.$$

Here $\theta_{i,\text{seen}}$ is the frozen 100k initialization. The penalty keeps trainable weights near the seen policy. The single deterministic λ was locked before evaluation and was not swept.

Method	Target	Retain	Δtarget	Δretain	Wall	VRAM
Naive	90.8%	20.6%	–	–	162.8s	7,540 MiB
Frozen-Stats	90.8%	21.7%	+0.0 pp	+1.1 pp	311.3s	7,540 MiB
L2-SP	**87.5%**	**31.7%**	**-3.3 pp**	**+11.1 pp**	297.3s	8,036 MiB

Table 4. Matched $N = 1$ comparison. Naive was reused, not rerun.

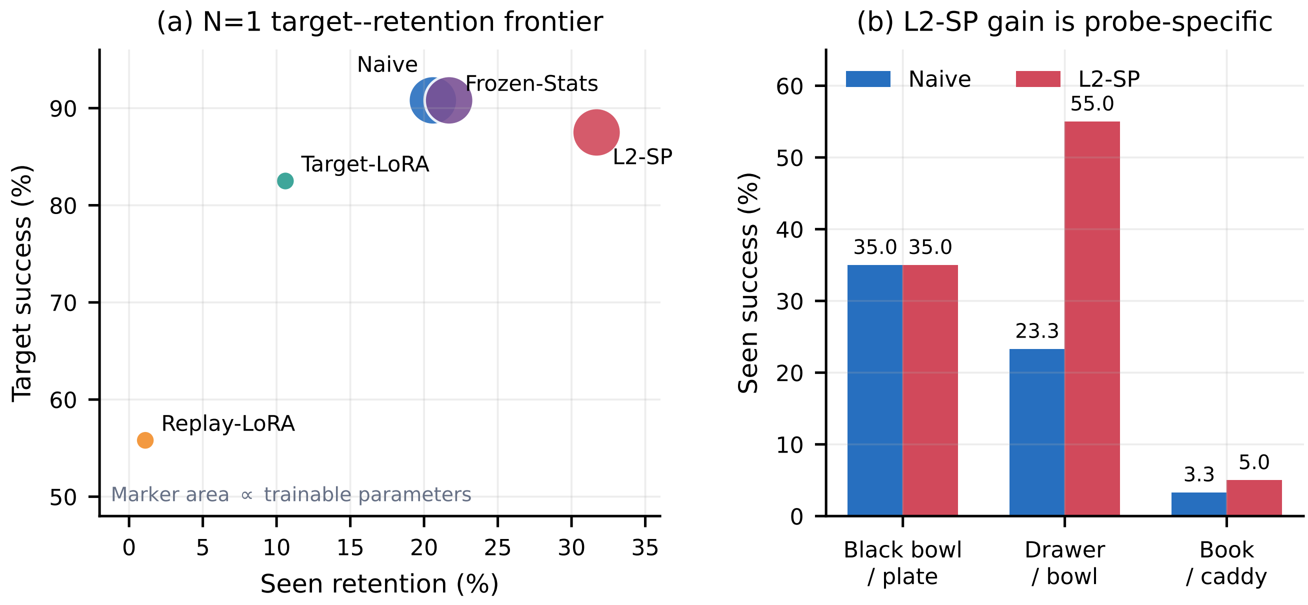
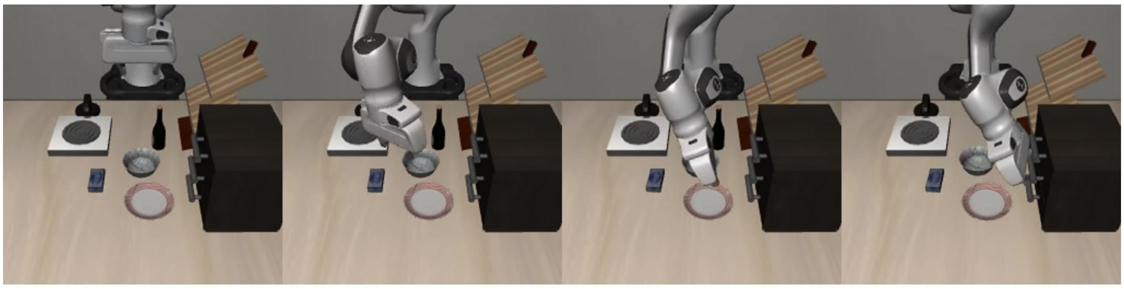


Figure 2. Updated $N = 1$ frontier and frozen-probe breakdown. Marker area in panel (a) is proportional to trainable parameters.

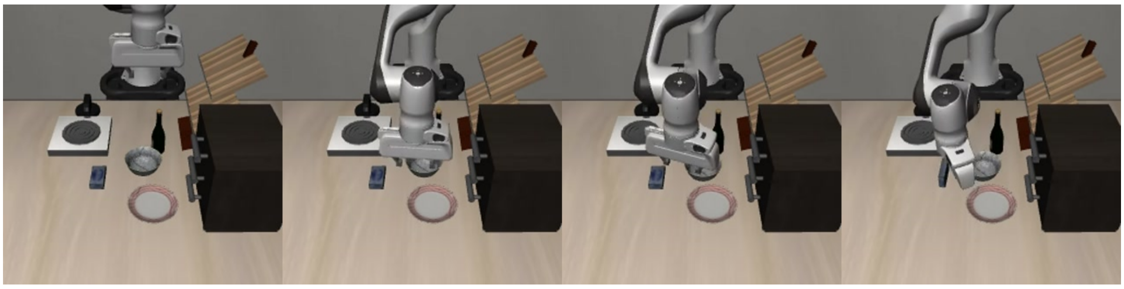
Frozen statistics alone change little. L2-SP gains 11.1 retention points for 3.3 target points. It is a useful Pareto trade-off, not strict dominance. Also, 19 of its 20 extra retained successes come from `drawer_bowl`. This supports anchoring, but not broad LIBERO-90 preservation. Appendix B.3 reports every task and seed.

7 Failures, predictions, and conclusion

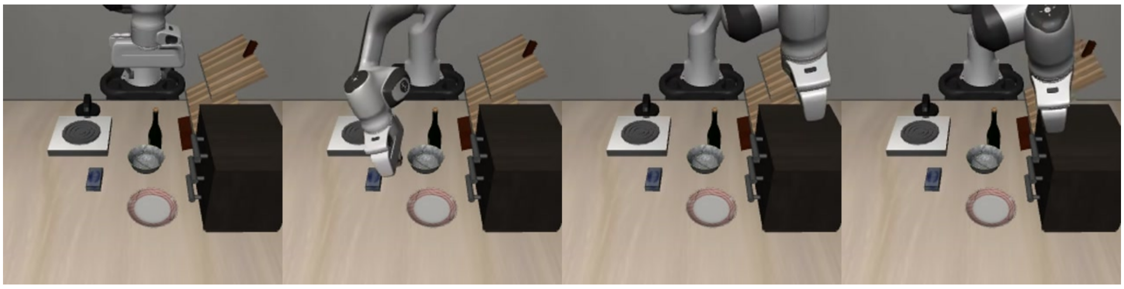
These are real failed zero-shot rollouts at seed 1000. Frames are at 0, 5, 10, and 15 seconds; no successful video was selected.



Drawer. The arm does not approach the middle drawer handle.



Bowl. The arm reaches the dish area but does not complete grasp and placement.



Wine. The first reach goes toward a distractor; the bottle is not picked up.

Separating tests. For Drawer, compare the original image with a handle crop or point marker. For Bowl, compare a normal start with the bowl already grasped. For Wine, cross an object-colour swap with a grasped start. These tests separate grounding, grasping, identity, and placement failures; details are in Appendix E.

Predictions versus results. I correctly expected the largest gain before five demos and later saturation. I incorrectly expected Replay-LoRA > Target-LoRA > naive. The actual order was the reverse. A small update can still interfere with old skills, and replay is unsafe when action scales are inconsistent.

Idea graveyard. I rejected target-task hyperparameter sweeps, target-overlay statistics for seen deployment, and full-VLM fine-tuning. These would tune on test tasks, confound retention, or add compute without testing the key question.

Most surprising result. One demo reaches 90.8% target success, but naive adaptation leaves 20.6% retention. Parameter efficiency does not solve this. Weight anchoring recovers part of the old behaviour: 31.7% retention at 87.5% target success.

The evidence is limited to three target tasks, two train seeds, and three retention probes. The next experiment should use a preregistered broader retention panel for the existing Naive and L2-SP checkpoints, then add train seeds. Future λ comparisons should use separate calibration tasks.

All 600 new rollouts were independently recounted. Hashes, demo IDs, seeds, and the canonical statistics digest match; `integrity_ok=true`. The full suite reports 304 passed, 8 skipped, and 0 failed. Exact protocols, tables, curves, hashes, and artifact paths follow in this same PDF.

Technical Appendix

One Shot Is All You Need. Or Is It?

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This appendix is part of the same source file and PDF as the four-page main report. It records the complete protocol, percentages and raw counts, training curves, compute, controls, hashes, and artifact map.

A Frozen protocol and provenance

A.1 Immutable sources

Item	Frozen value
Base model	`lerobot/smolvla_base`, revision c83c3163b8ca9b7e67c509fffd9121e66cb96205
Dataset	`nvidia/LIBERO_LeRobot_v3`, revision e5907374380b8f96511957e6ba5582be52a1e179
Seen data	Full available `libero_90`: 3,921 episodes, 569,249 frames, 73 instruction rows
Seen checkpoint	`step_100000`; weights SHA-256 2cd510a594a87580f7368b782ca9b37332c0e5002d807093c759e95fbfb57c88
Seen statistics	`libero_90`; digest b159b6fed3e52edf25bd39b377dd64940221b7a030362daf7f726b1c2ecb30cf
First $N = 1$ episodes	Drawer 20; Bowl 13; Wine 6
Train seeds	42 and 123
Target evaluation	Seeds 1000–1019; 20 rollouts per checkpoint
Retention	`black_bowl_plate`, `drawer_bowl`, `book_caddy`; seeds 1000–1009; canonical `libero_90` statistics
Environment	Hard reset, relative control, 300-step horizon; predict 50 actions and execute 10

Table A1. Frozen data, checkpoint, and evaluation protocol.

Target tasks are held out by exact instruction text: *open the middle drawer of the cabinet*, *put the bowl on the stove*, and *put the wine bottle on top of the cabinet*. They are not selected by dataset row number.

A.2 Training settings

The vision encoder and VLM backbone stay frozen. Naive, Frozen-Stats, and L2-SP train the Action Expert plus state/action projections: exactly 99,880,992 parameters. Target-LoRA and Replay-LoRA use rank $r = 64$, $\alpha = 64$, zero dropout, plus projections: 4,215,632 parameters.

Stage	Learning rate	Maximum	Stop rule	Batch	Precision
Seen expert	10^{-4}	100,000	fixed	32	bf16
Naive target	10^{-4}	12,000	min(100 epochs, cap)	32	bf16
Target-LoRA	10^{-3}	12,000	min(100 epochs, cap)	32	bf16
Replay-LoRA	10^{-3}	12,000	min(100 epochs, cap)	32	bf16
Frozen-Stats	10^{-4}	12,000	min(100 epochs, cap)	32	bf16
L2-SP	10^{-4}	12,000	same + $\lambda = 0.01$	32	bf16

Table A2. Training recipes. No method uses success-based stopping or reruns.

All runs use AdamW, weight decay 10^{-2}, betas $(0.9, 0.95)$, gradient clipping at 1.0, 1,000-step warm-up, and cosine decay. Replay draws 75% target and 25% `libero_90`. Frozen-Stats and L2-SP use only the first target demo and canonical seen statistics. L2-SP computes the raw-sum anchor penalty in FP32 against the frozen seen initialization. The gripper conversion is $a_{\text{env}} = 1 - 2a_{\text{data}}$. Six stored demonstrations replay with 100.0% (6/6) success. The matched preflight found finite normalized actions and gripper range $[-1.059968, 0.943425]$.

B Complete evaluation results

B.1 Naive target cost curve

N	Drawer 42	Drawer 123	Bowl 42	Bowl 123	Wine 42	Wine 123	Pooled
0	5.0% (1/20)		0.0% (0/20)		0.0% (0/20)		1.7% (1/60)
1	85.0% (17/20)	95.0% (19/20)	100.0% (20/20)	100.0% (20/20)	80.0% (16/20)	85.0% (17/20)	90.8% (109/120)
2	50.0% (10/20)	100.0% (20/20)	100.0% (20/20)	90.0% (18/20)	80.0% (16/20)	80.0% (16/20)	83.3% (100/120)
5	85.0% (17/20)	70.0% (14/20)	100.0% (20/20)	100.0% (20/20)	85.0% (17/20)	95.0% (19/20)	89.2% (107/120)
10	95.0% (19/20)	100.0% (20/20)	100.0% (20/20)	100.0% (20/20)	95.0% (19/20)	90.0% (18/20)	96.7% (116/120)
25	100.0% (20/20)	100.0% (20/20)	100.0% (20/20)	100.0% (20/20)	80.0% (16/20)	90.0% (18/20)	95.0% (114/120)

Table A3. All naive target cells. Every success entry gives percentage and raw count.

N	Target success	95% Wilson interval	Change from zero shot
0	1.7% (1/60)	[0.3, 8.9]%	–
1	90.8% (109/120)	[84.3, 94.8]%	+89.2 pp
2	83.3% (100/120)	[75.7, 88.9]%	+81.7 pp
5	89.2% (107/120)	[82.3, 93.6]%	+87.5 pp
10	96.7% (116/120)	[91.7, 98.7]%	+95.0 pp
25	95.0% (114/120)	[89.5, 97.7]%	+93.3 pp

Table A4. Pooled target curve with uncertainty.

B.2 Corrected seen retention

N	Drawer-trained	Bowl-trained	Wine-trained	Pooled	Change
0	Frozen reference 80.0% (24/30)			80.0% (24/30)	–
1	10.0% (6/60)	33.3% (20/60)	18.3% (11/60)	20.6% (37/180)	-59.4 pp
2	8.3% (5/60)	21.7% (13/60)	5.0% (3/60)	11.7% (21/180)	-68.3 pp
5	0.0% (0/60)	0.0% (0/60)	0.0% (0/60)	0.0% (0/180)	-80.0 pp
10	0.0% (0/60)	0.0% (0/60)	0.0% (0/60)	0.0% (0/180)	-80.0 pp
25	0.0% (0/60)	0.0% (0/60)	0.0% (0/60)	0.0% (0/180)	-80.0 pp

Table A5. Corrected retention with canonical `libero_90` statistics.

Across the 30 adapted naive cells, target success and corrected retention have Pearson $r = 0.026$. This is descriptive because cells share tasks.

B.3 All matched $N = 1$ cells

Target uses 20 rollouts. Retention uses 30 rollouts: three probes and ten fixed seeds. All success cells give percentage and raw count.

Table A6: Every matched $N = 1$ task, seed, result, wall time, and peak VRAM.

Method	Target task	Seed	Target	Retention	Train wall	Peak VRAM
Naive	Drawer	42	85.0% (17/20)	13.3% (4/30)	199.1s	7,540 MiB
Naive	Drawer	123	95.0% (19/20)	6.7% (2/30)	199.4s	7,540 MiB
Naive	Bowl	42	100.0% (20/20)	33.3% (10/30)	162.2s	7,540 MiB
Naive	Bowl	123	100.0% (20/20)	33.3% (10/30)	162.0s	7,540 MiB
Naive	Wine	42	80.0% (16/20)	20.0% (6/30)	127.2s	7,540 MiB
Naive	Wine	123	85.0% (17/20)	16.7% (5/30)	126.9s	7,540 MiB
Target-LoRA	Drawer	42	75.0% (15/20)	0.0% (0/30)	314.7s	6,944 MiB
Target-LoRA	Drawer	123	40.0% (8/20)	3.3% (1/30)	315.0s	6,938 MiB
Target-LoRA	Bowl	42	100.0% (20/20)	30.0% (9/30)	229.7s	6,944 MiB
Target-LoRA	Bowl	123	100.0% (20/20)	23.3% (7/30)	229.3s	6,944 MiB
Target-LoRA	Wine	42	95.0% (19/20)	3.3% (1/30)	187.0s	6,882 MiB
Target-LoRA	Wine	123	85.0% (17/20)	3.3% (1/30)	188.2s	6,944 MiB
Replay-LoRA	Drawer	42	15.0% (3/20)	0.0% (0/30)	435.9s	6,944 MiB
Replay-LoRA	Drawer	123	15.0% (3/20)	0.0% (0/30)	436.0s	6,944 MiB
Replay-LoRA	Bowl	42	100.0% (20/20)	3.3% (1/30)	332.3s	6,944 MiB
Replay-LoRA	Bowl	123	100.0% (20/20)	3.3% (1/30)	332.3s	6,944 MiB
Replay-LoRA	Wine	42	40.0% (8/20)	0.0% (0/30)	159.7s	6,944 MiB
Replay-LoRA	Wine	123	65.0% (13/20)	0.0% (0/30)	159.9s	6,944 MiB
Frozen-Stats	Drawer	42	100.0% (20/20)	13.3% (4/30)	378.3s	7,540 MiB
Frozen-Stats	Drawer	123	100.0% (20/20)	13.3% (4/30)	376.9s	7,540 MiB
Frozen-Stats	Bowl	42	100.0% (20/20)	13.3% (4/30)	326.1s	7,540 MiB
Frozen-Stats	Bowl	123	100.0% (20/20)	16.7% (5/30)	325.5s	7,540 MiB
Frozen-Stats	Wine	42	75.0% (15/20)	40.0% (12/30)	230.4s	7,540 MiB
Frozen-Stats	Wine	123	70.0% (14/20)	33.3% (10/30)	230.4s	7,540 MiB
L2-SP	Drawer	42	100.0% (20/20)	23.3% (7/30)	392.3s	8,036 MiB
L2-SP	Drawer	123	100.0% (20/20)	23.3% (7/30)	392.3s	8,036 MiB
L2-SP	Bowl	42	95.0% (19/20)	23.3% (7/30)	314.8s	8,036 MiB
L2-SP	Bowl	123	85.0% (17/20)	33.3% (10/30)	314.7s	8,036 MiB
L2-SP	Wine	42	70.0% (14/20)	40.0% (12/30)	185.2s	8,036 MiB
L2-SP	Wine	123	75.0% (15/20)	46.7% (14/30)	184.5s	8,036 MiB

B.4 Aggregate intervals and probe totals

Method	Target	95% CI	Retention	95% CI
Naive	90.8% (109/120)	[84.3,94.8]%	20.6% (37/180)	[15.3,27.0]%
Target-LoRA	82.5% (99/120)	[74.7,88.3]%	10.6% (19/180)	[6.9,15.9]%
Replay-LoRA	55.8% (67/120)	[46.9,64.4]%	1.1% (2/180)	[0.3,4.0]%
Frozen-Stats	90.8% (109/120)	[84.3,94.8]%	21.7% (39/180)	[16.3,28.2]%
L2-SP	87.5% (105/120)	[80.4,92.3]%	31.7% (57/180)	[25.3,38.8]%

Table A7. Matched $N = 1$ aggregate results and Wilson intervals.

Method	Black bowl / plate	Drawer / bowl	Book / caddy
Naive	35.0% (21/60)	23.3% (14/60)	3.3% (2/60)
Target-LoRA	20.0% (12/60)	11.7% (7/60)	0.0% (0/60)
Replay-LoRA	3.3% (2/60)	0.0% (0/60)	0.0% (0/60)
Frozen-Stats	25.0% (15/60)	36.7% (22/60)	3.3% (2/60)
L2-SP	35.0% (21/60)	55.0% (33/60)	5.0% (3/60)

Table A8. Retention totals by frozen probe.

C Training diagnostics

C.1 Seen expert

The 100k run takes 38,870 seconds (10 h 47 min 50 s), processes 3.2M samples, reserves 7,536 MiB at the final logged step, and ends at loss 0.23494.

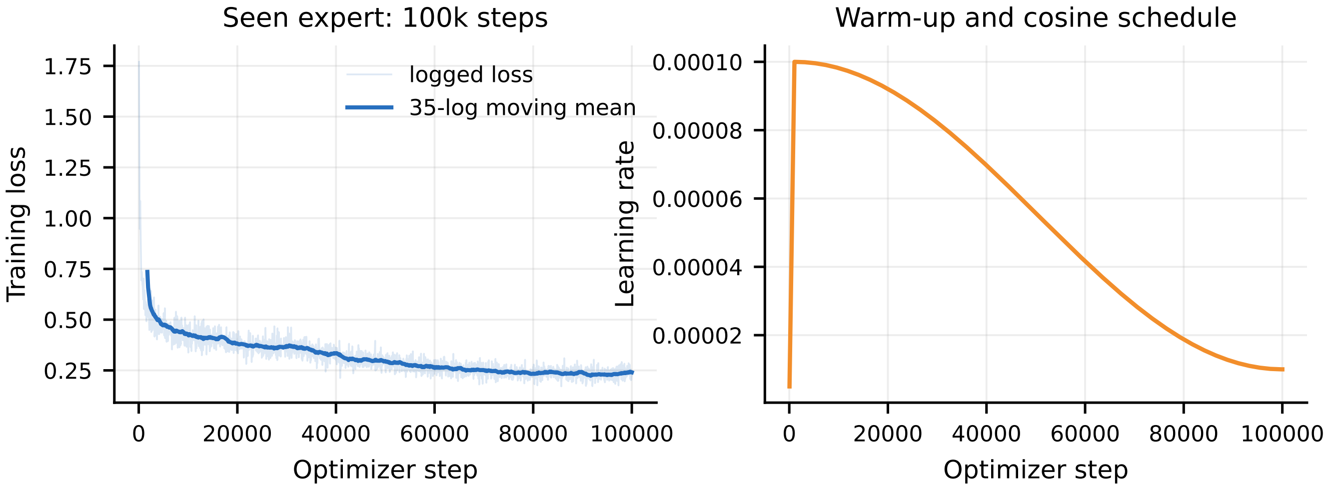

Figure A1. Seen-expert loss, moving mean, warm-up, and cosine schedule.

C.2 Naive target training

The stop rule is 100 dataset epochs capped at 12k steps. At $N = 1$, final steps are 500 for Drawer, 400 for Bowl, and 300 for Wine.

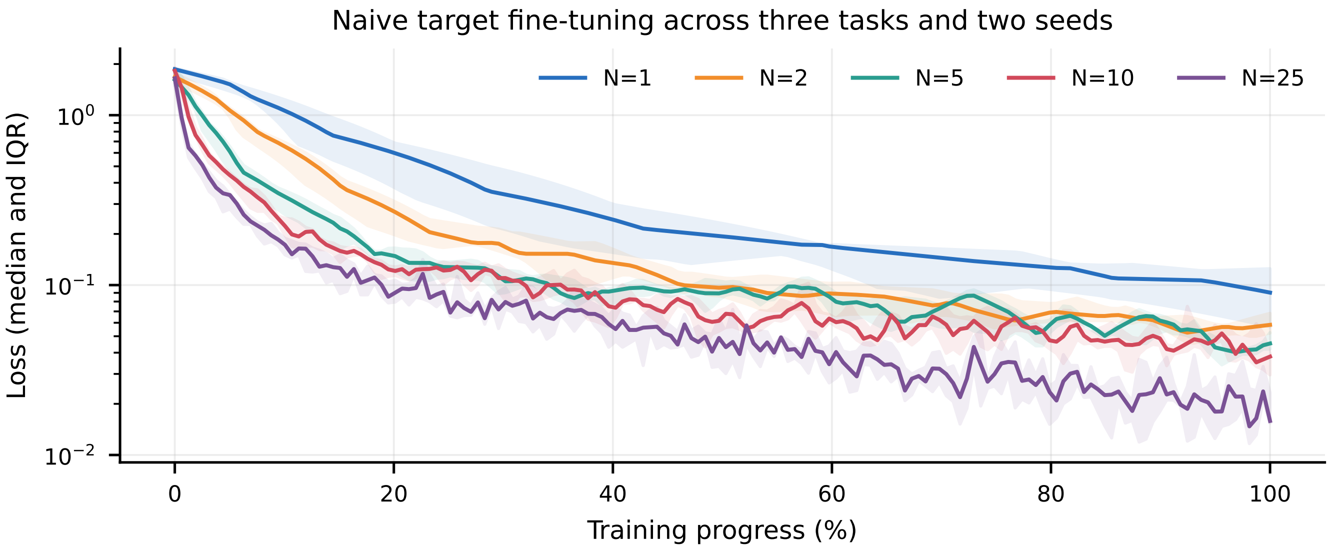

Figure A2. Naive target loss for all budgets. Curves show the median and interquartile range across three tasks and two seeds.

C.3 LoRA and replay diagnostics

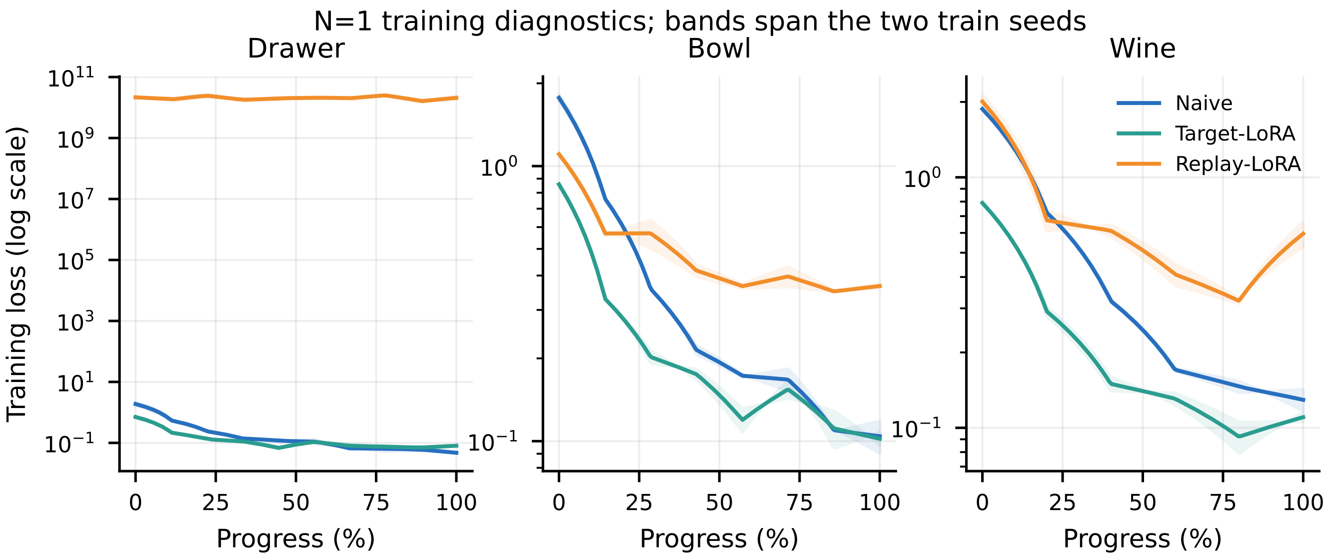

Figure A3. $N = 1$ training by task for Naive, Target-LoRA, and Replay-LoRA.

Replay-LoRA Drawer is numerically pathological. The first target demo has gripper standard deviation 10^{-6}, while seen replay has both states. Target-only normalization creates extreme replay targets. Final Drawer losses are 1.44×10^{10} and 2.72×10^{10}. Bowl and Wine do not show the same explosion, so this does not explain every lost seen success.

C.4 Frozen-Stats and L2-SP diagnostics

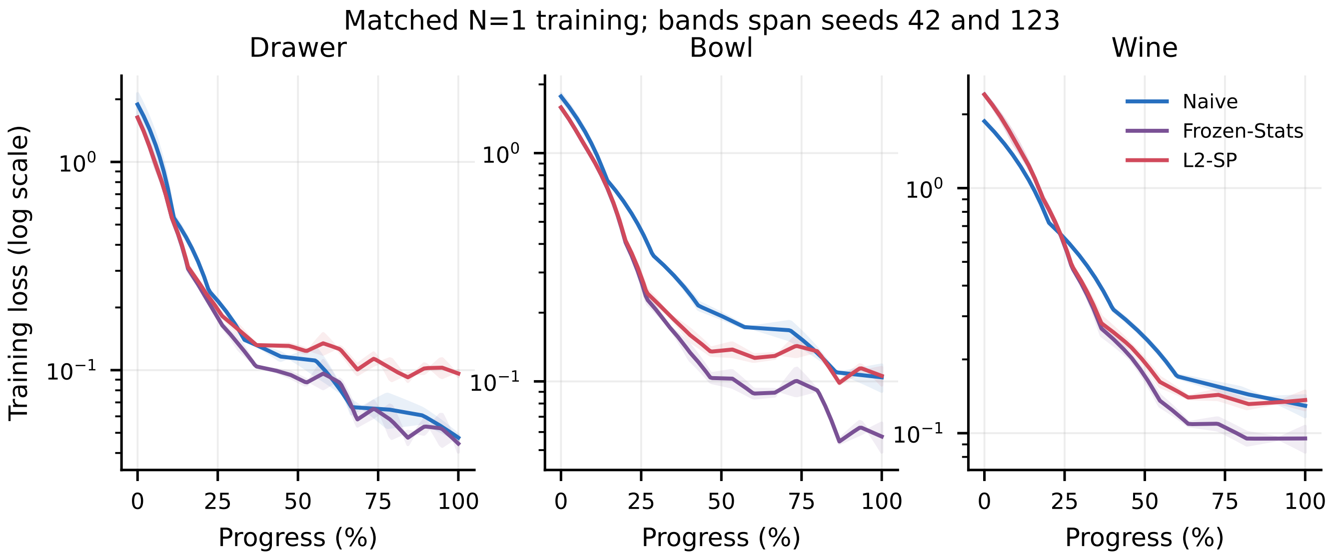

Figure A4. Matched $N = 1$ loss curves. L2-SP includes the fixed anchor penalty; bands span seeds 42 and 123.

Method	Cell	Final step	Samples	Final logged loss
Frozen-Stats	Drawer 42 / 123	500	16,000	0.03646 / 0.05225
Frozen-Stats	Bowl 42 / 123	400	12,800	0.04071 / 0.07398
Frozen-Stats	Wine 42 / 123	300	9,600	0.07232 / 0.11800
L2-SP	Drawer 42 / 123	500	16,000	0.09008 / 0.10256
L2-SP	Bowl 42 / 123	400	12,800	0.08710 / 0.12436
L2-SP	Wine 42 / 123	300	9,600	0.11272 / 0.16037

Table A9. Exact final records for the matched training cells.

C.5 Compute, concurrency, and end-to-end time

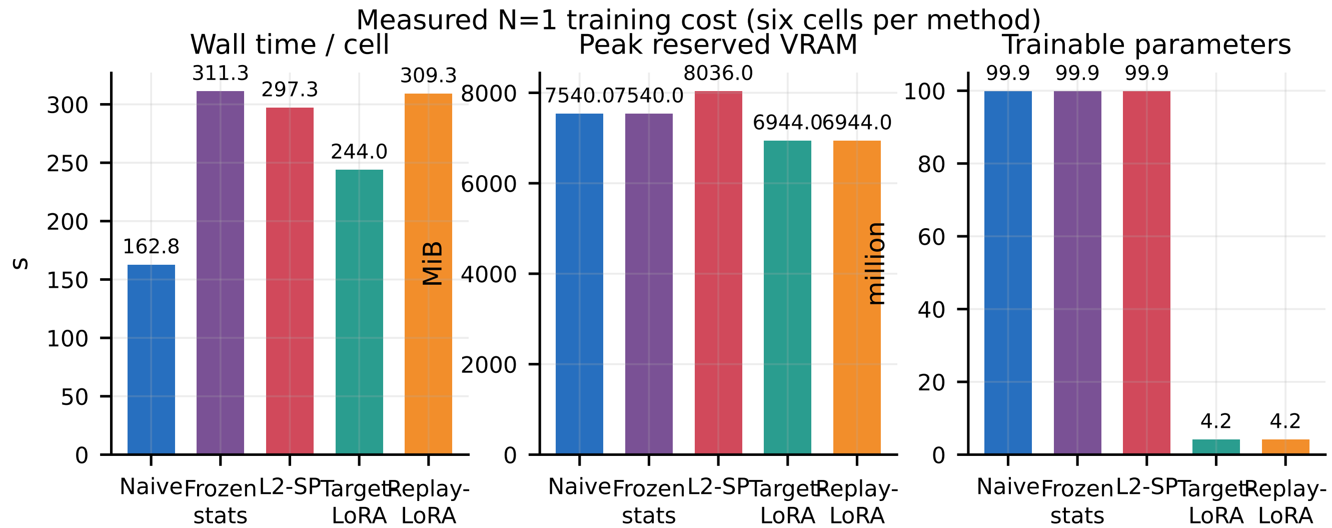

Figure A5. Mean wall time per cell, peak reserved VRAM, and trainable parameters.

For the matched grid, two jobs reach 194.0 aggregate samples/s and four reach 202.7 samples/s (+4.4%). Four-way concurrency was safe and reserved about 33–35 GiB of the 97.9 GiB GPU.

The matched 12-cell grid took 47 min 39 s: training 15 min 57 s, target evaluation 8 min 58 s, and retention 22 min 44 s. Including action audit, smokes, and concurrency checks, the run took about 58 minutes. There were no failed stages.

D Controls and integrity

D.1 Language control

Task	Correct instruction	Wrong instruction	Mean action L_2
Drawer	5.0% (1/20)	0.0% (0/20)	0.5820
Bowl	0.0% (0/20)	0.0% (0/20)	0.9982
Wine	0.0% (0/20)	0.0% (0/20)	0.9661

Table A10. Paired control on identical initial-state fingerprints.

Mean cosine gaps are 0.3273, 0.7731, and 0.6626. Changed trajectories show language sensitivity, but low success does not prove strong zero-shot instruction understanding.

D.2 Normalization 2-by-2 control

Weights	Statistics	Success	Interpretation
Frozen seen	`libero_90`	80.0% (12/15)	expected competence
Frozen seen	target overlay	0.0% (0/60)	statistics break deployment
Adapted target	target overlay	0.0% (0/60)	confounded old result
Adapted target	`libero_90`	10.0% (6/60)	corrected weight test

Table A11. Control separating weight change from statistics change.

The earlier 0.0% (0/900) result is kept only as a deployment warning, not as evidence of catastrophic forgetting.

D.3 Matched-run integrity

All 12 matched checkpoints have distinct final weight hashes. The 240 target and 360 retention records were independently recounted from raw JSONL: Frozen-Stats gives 90.8% (109/120) and 21.7% (39/180); L2-SP gives 87.5% (105/120) and 31.7% (57/180). Every record matches the registered task, demo episode, train seed, evaluation seed, checkpoint hash, and canonical statistics digest. The driver reports `integrity_ok=true` and zero failed stages.

Implementation commit: f77c469c5f90a2b7bba37988b39848b0b7101abe.

Results commit: 66f04e9beaaa74e074e6aecb49ffb47bfd7bc3c6.

The full suite reports 304 passed, 8 skipped, and 0 failed.

E Failure hypotheses and separating tests

Drawer: spatial grounding or low-level control? Compare the original image, a handle crop, and a point-marked image while keeping state and instruction fixed. Measure end-effector distance to the middle handle before contact. Improvement only with visual guidance supports a grounding failure.

Bowl: grasping or relational planning? Use two matched starts: the normal scene and the same scene with the bowl already in the gripper. Success only from the second start isolates grasping; failure in both supports a placement or long-horizon problem.

Wine: object identity or placement horizon? Cross two interventions: swap bottle and distractor colours, and start with the bottle already grasped. A reach that follows colour supports an identity error; failure from the grasped start points to cabinet-top placement.

F Artifact map and next experiment

Evidence	Location
Unified source	`report/latex/paper.tex`
Compiled paper and appendix	`report/latex/build/paper.pdf`
Frozen predictions	`predictions.md`
Naive $N = 1, 2$	`/mnt/vla/validation/TODO28_n12/results.md`
Corrected naive retention	`/mnt/vla/validation/TODO28_retention_libero90/retention.md`
LoRA result	`report/tables/task2_n1_results.md`
Frozen-Stats/L2-SP	`report/tables/frozen_stats_l2sp_n1_results.md`
Matched summary	`/mnt/vla/validation/TODO32/summary.json`
Matched checkpoints	`/mnt/vla/runs/target_matched_n1`
Matched target evaluation	`/mnt/vla/eval/target_matched_n1`
Matched retention	`/mnt/vla/eval/seen_retention_libero90_matched_n1`

Table A12. Source and evidence locations. Runtime paths are on the experiment host; portable summaries are committed.

The next locked experiment should evaluate existing Naive and L2-SP checkpoints on a broader, preregistered seen-task panel. This tests whether the gain extends beyond `drawer_bowl`. If it does, add train seeds. Any future λ values should be chosen on separate calibration tasks. Useful mechanistic follow-ups are parameter drift by Action Expert component and a matched function-space distillation penalty.

G References

References

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